Let the Fixture Carry Precision in Learned Chains

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Abstract: A learned placement policy that is chained compounds its own error, so a transfer gap that doubles the policy's per-stage error cuts the usable horizon of the chain by about four. We study a classical way to make chain length insensitive to that gap: after every placement, push the part against a physical reference until contact fixes its pose. A funnel model makes the trade explicit. Chain failure splits into the policy's error mass beyond the funnel's capture radius, which is centimetres wide, and the seating residual of the contact step, which is set by contact physics and fixture geometry, not by the policy. On a simulated Panda line-up with a diffusion policy, four matched arms show that the gain is contact, not a robot reset or extra motion (restore vs. a motion-matched control, 17:0 discordant chains, $p = 1.5 \times 10^{-5}$). Sweeping the policy-side gap and the contact model then shows where each transfer factor lands: the restored chain succeeds in 40 of 40 chains across an 8× range of policy scatter and a 2 cm calibration bias, while unchanged restore code breaks only when fixture friction rises. That failure is observable on board, from the plate position at which the push stalls.

Keywords: sim-to-real transfer, long-horizon manipulation, contact funnels

1 Introduction

Long-horizon manipulation is often built by chaining one learned skill. When later targets are defined relative to earlier parts, as in assembly, stacking and kitting, a 3 mm error at stage k moves the target of stage $k + 1$, and nothing in the chain pulls the state back. This makes chains unusually fragile under transfer. Every source of sim-to-real gap that degrades a single placement (camera calibration, lighting, object mass, friction) enters the chain through the policy's per-stage error σ, and for a compounding chain the horizon at which half the chains still succeed scales as $(\tau/\sigma)^2$ for a tolerance τ (Prop. 1). A policy that is twice as sloppy in the field as in simulation keeps a quarter of its horizon.

Manipulation has had a remedy for compounding pose error since the 1980s: push the part against something. A constraining contact maps a whole region of uncertain poses onto one seated pose, which is Mason's funnel [1], preimage backchaining [2] and sensorless orienting [3, 4]. We ask what such a step does to the transfer problem when it is inserted between the stages of a learned chain, and argue that it changes which quantities have to transfer: the policy's errors matter only beyond a centimetre-scale capture radius, and the millimetre-scale residual becomes a property of contact physics and fixture geometry that can be identified and checked on board.

Contributions. (1) A funnel model of a chained policy that splits chain failure into a policy term evaluated at the capture radius and a contact term evaluated at the tolerance (Sec. 2). (2) A four-arm protocol, run with a learned diffusion policy on two task topologies, that

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separates contact from the robot reset and extra motion any recovery step also adds (Sec. 4). (3) Transfer sweeps over the policy's scatter and bias, the and the contact model, which measure each term of the split (Sec. 5). (4) An on-board check of the contact term from proprioception alone, and the field tests the model says are needed (Sec. 6).

Related work. Sim-to-real work usually attacks the policy side of the gap, by randomizing appearance or dynamics so that σ stays small after transfer [5, 6], and skill-chaining work regularizes training so that one skill's terminal states fall inside the next skill's initiation set [7, 8]. Both change the policy. Classical funnels [1, 2, 3, 9, 4] and sequential composition [10] change the state instead, and manufacturing models how fixturing stops variation from propagating through multistage assembly [11, 12]; those analyses assume a mechanics model that certifies capture. BATON [13] restores the robot between learned subtasks with non-contact motions. We measure what restoring the part adds, and what it does to the transfer question.

2 A Funnel Model of Chain Transfer

Let D_k be the signed deviation of part k from its nominal pose after stage k, with $D_0 = 0$, and let the policy land part k with error ξ_k relative to its target, i.i.d. with distribution F_σ whose scale σ is what transfer degrades. A part succeeds if $|D_k| \le \tau$. In the bare line-up the target is defined by the predecessor, so $D_k = D_{k-1} + \xi_k$.

Proposition 1 (bare chains compound). If $\xi_k \sim \mathcal{N}(0, \sigma^2)$, the final part succeeds with probability $2\Phi(\tau/\sigma\sqrt{N}) - 1$, an upper bound on chain success, which falls to one half at $N_{1/2} = (\tau/0.674\,\sigma)^2$. The horizon's elasticity in the transfer gap is $\partial \log N_{1/2}/\partial \log \sigma = -2$.

A contact step behaves differently. Model it as a funnel with capture radius R and seating residual η: if the landing deviation $|D_{k-1}+\xi_k|$ (line-up) or $|\xi_k|$ (fixed cells) is at most R, the step leaves $D_k = \eta_k$, an i.i.d. draw from the contact step's residual distribution G; otherwise the stage is lost.

Proposition 2 (the funnel splits the transfer burden). If $R > \tau$, the funnel acts at every stage and later stages do not move earlier parts, chain failure is at most $N\big[\,\Pr_{F_\sigma}(|\xi| > R - \tau) + \Pr_G(|\eta| > \tau)\,\big]$.

The first term is the policy's transfer gap, but read at $R - \tau$ rather than at τ, and the second term does not involve the policy at all. For a Gaussian policy and $R = 5\tau$ the first term at $\sigma = \tau$ is 6×10^{-5} per stage: within the capture radius the horizon's elasticity in σ is near zero, and the chain inherits the transfer properties of the contact step instead (G from friction, stiffness and fixture; R from push geometry). Two corollaries follow (App. A): a step that never moves the part, however long, leaves the law of $(D_1, \ldots, D_N)$ unchanged unless it changes the robot state the next placement starts from; and a policy bias b (a calibration offset) is removed entirely while bias plus scatter stays within $R - \tau$.

3 Setup

A Franka Panda in robosuite [14] (MuJoCo) places 4 cm cubes in a chain of N stages. In the line-up, stage k targets the actual resting pose of cube $k-1$; in 3×4 packing, each cube targets a fixed cell of a zero-clearance grid. The learned placer is a robomimic diffusion policy [15, 16] trained by behavior cloning on 150–300 scripted demonstrations and sampled with 10 DDIM steps. The transfer sweeps use a scripted placer with injected error, so that σ, b and the physics can be set exactly. Success means every cube is within $\tau = 8$ mm of its nominal cell at the end of the chain.

Arms. All arms share seeds and placements. No reset: nothing happens between stages. Home: the end effector returns to its pre-placement pose without touching anything, a

Table 1: Learned diffusion policy, chain success at $\tau = 8$ mm (line-up 10 paired seeds, packing 6). In packing every arm homes the robot. Right: within-stage contraction, the median ratio of a part's deviation after the post-placement step to its deviation at landing.

	3-D line-up			3×4 packing			contraction
Arm	N=2	N=4	N=8	N=4	N=8	N=12	line-up / pack
No reset	0/10	0/10	0/10	–	–	–	1.00 / –
Home	7/10	0/10	0/10	0/6	0/6	0/6	1.00 / 1.00
No-op	9/10	0/10	0/10	0/6	0/6	0/6	1.00 / 1.00
Restore	10/10	10/10	6/10	6/6	4/6	0/6	0.21 / 0.39

robot-only reset of the kind used between VLA subtasks [13]. No-op: home plus a no-contact motion near the part of the same duration as the restore. Restore: the closed gripper, used as a plate, pushes the part against its references one axis at a time, then homes. Pushes are saturated and stall-terminated, since a proportional controller's force vanishes exactly where seating needs it; references are found from collision geometry (the nearest static or seated face in $-x$ and $-y$), so one routine serves both topologies.

4 Contact, Not Reset or Time

Table 1 gives the learned-policy results. Resetting the robot helps (home vs. no reset, 7:0 discordant chains, exact McNemar $p = 0.016$), in line with BATON's ablation [13], but every non-contact arm fails all ten line-up chains by $N = 4$. Extra motion near the part buys nothing measurable (no-op vs. home, 2:0, $p = 0.5$; packing 0:0), as the no-contact corollary predicts. Contact separates the arms: restore beats the motion-matched control 17:0 on the line-up ($p = 1.5 \times 10^{-5}$) and 16:0 in packing ($p = 3 \times 10^{-5}$). The mechanism shows per stage: every non-contact arm passes the landing error on unchanged (contraction 1.00), while restore leaves 0.21 of it on the line-up and 0.39 in packing (bare line-up deviation grows from 3.1 to 155.6 mm over 8 stages; restore stays at 1.2–12.0 mm). Restore costs 2.3× the steps of a bare placement.

5 Which Side of the Split Does Each Transfer Factor Land On?

Policy-side gap. We degrade the scripted placer as transfer would: Gaussian scatter σ from 2 to 16 mm, and a systematic bias b (a calibration offset) of 5–20 mm at $\sigma = 4$ mm, on the $N = 8$ line-up with 5 paired seeds per setting. The bare chain fails every chain at every setting, and its worst cube grows with the gap (median 63 mm at $\sigma = 2$ mm, 100 mm at $\sigma = 16$ mm, 208 mm at $b = 20$ mm). Restore succeeds in 40 of 40 chains at $\tau = 8$ mm across all eight settings, with a median worst cube of 1.8–5.0 mm and a within-stage contraction of 0.07–0.23. The restored chain's error is flat in an 8× range of σ and in a bias of half a cube width, as Prop. 2 predicts inside the capture radius; only the tighter 4 mm tolerance starts to show the gap (5/5 chains at $\sigma = 4$ mm, 1–2/5 at $\sigma \geq 12$ mm or $b \geq 10$ mm).

Contact-model shift. The second term of the split belongs to the contact step, so we keep the restore routine exactly as tuned in robosuite's default physics and execute it under shifted contact models: contact time constant 0.005–0.04 s, sliding friction 0.5–2×, pyramidal instead of elliptic friction cones, and impedance ratio 1 instead of 20 ($N = 4$, $\sigma = 4$ mm, 2 paired seeds per setting; the run was cut short by memory on our machine, so these are pilot-scale). Figure 1b shows the result. Most shifts leave the funnel intact (median worst cube 0.1–2.8 mm, all chains successful). Friction does not: at 1.5× and 2× the contraction rises to 0.67 and 0.99, so the step stops removing error, and the worst cube reaches 11.7 and 13.6 mm. Very stiff contact costs less (5.8 mm); the bare chain fails everywhere. The

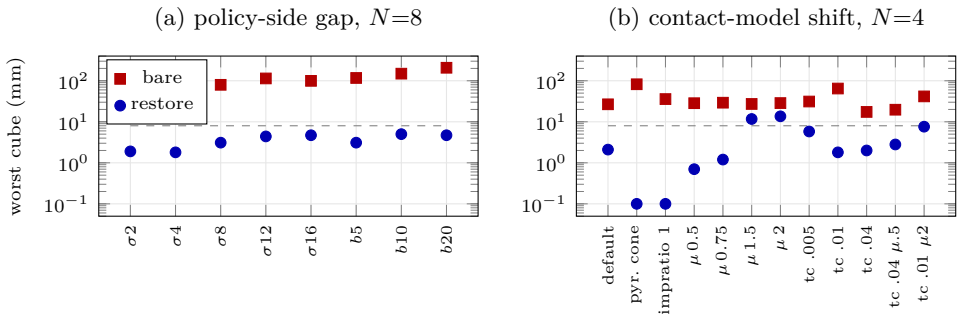


Figure 1: Median worst-cube deviation per setting (log scale; dashed line: $\tau = 8$ mm). (a) Degrading the placer (scatter σ and bias b, mm) moves the bare chain but not the restored one. (b) Executing the unchanged restore under shifted contact physics: only high friction and very stiff contact move it toward the tolerance.

burden has moved, not vanished: to the fixture's friction, which a field team can measure with a pull test.

On-board seat check. The contact term can also be observed at deployment. Where the plate stalls at the end of the final push, read from proprioception and corrected by one offset calibrated in default physics, estimates the seated part's deviation to a median of 0.3–1.7 mm (90th percentile 1.0–4.2 mm) over 260 restore stages of the gap sweep, abstaining when the part was already seated and the plate never reached it. Under the shifted contact models, where the offset was not recalibrated, the error grows to 0.3–4.8 mm, and flagging a stage when the estimate exceeds τ catches 7 of the 11 stages that actually ended outside tolerance, with 5 false alarms in 64 stages. The failure friction introduces, a push arrested short of its fence, is one the arm can feel.

6 Toward the Field

Read as a recipe for transfer, the split says what to validate on hardware and what to stop worrying about. Measure the funnel, not the policy: the capture radius R (here at least 2 cm, since biases of 20 mm and scatter of 16 mm were absorbed) and the seating residual G on the real fixture, which needs only scripted pushes of deliberately misplaced parts, no learned policy. Identify friction first: it is the one contact parameter in our sweep that broke the funnel, and the one most likely to differ between a simulated and a real fixture (surface finish, dust, wear). Monitor the stall: a plate that stops short of where the fixture says it should is an on-board flag of a failed seat, and the place to trigger a retry. Spend randomization on the policy only up to R: once its errors are inside the capture radius, further policy-side robustness buys no chain length.

Limitations. Every number is from MuJoCo, with cubes only; we have not measured R or G on hardware. The contact sweep is pilot-scale (2 seeds per setting). The restore routine reads part poses from the simulator to plan its approach, so perception error enters only through that approach. In zero-clearance packing, late pushes shift the finished block by 2–35 mm, so restore fails all chains at $N = 12$: the model's assumption that stages do not disturb each other is where it breaks.

Conclusion. Contact re-seating lets the policy be centimetre-accurate and takes the millimetres from the fixture; the next step is measuring a real fixture's capture radius and seating residual.

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A Proofs

Proposition 1. Unrolling $D_k = D_{k-1} + \xi_k$ from $D_0 = 0$ gives $D_N = \sum_{k \le N} \xi_k \sim \mathcal{N}(0, N\sigma^2)$, so $\Pr(|D_N| \le \tau) = 2\Phi(\tau/\sigma\sqrt{N}) - 1$. Chain success requires the final part to succeed, which bounds it above. Since $2\Phi(x) - 1 = \frac{1}{2}$ at $x \approx 0.674$, the bound is one half at $N_{1/2} = (\tau/0.674\sigma)^2$, and $\log N_{1/2} = 2\log\tau - 2\log(0.674\sigma)$ has derivative -2 in $\log\sigma$.

Proposition 2. Let $A_k = \{|\xi_k| > R - \tau\}$ and $B_k = \{|\eta_k| > \tau\}$. On the complement of $\bigcup_k (A_k \cup B_k)$ every part succeeds, by induction on k: if $|D_{k-1}| \le \tau$ (true for $D_0 = 0$), the landing deviation is at most $|D_{k-1}| + |\xi_k| \le \tau + (R - \tau) = R$ (and $|\xi_k| \le R$ when targets are fixed cells), so the funnel captures the part and leaves $D_k = \eta_k$ with $|\eta_k| \le \tau$. The model assumes later stages do not move earlier parts, so the final configuration is the per-stage one. A union bound over the $2N$ events gives failure at most $N[\Pr(|\xi| > R - \tau) + \Pr(|\eta| > \tau)]$. For $\xi \sim \mathcal{N}(0, \sigma^2)$, $R = 5\tau$ and $\sigma = \tau$, $\Pr(|\xi| > 4\tau) = 2(1 - \Phi(4)) \approx 6.3 \times 10^{-5}$.

No-contact corollary. If the post-placement step is the identity on part poses, D_k is the same function of $(\xi_1, \ldots, \xi_k)$ as in the bare chain; if in addition the step leaves the law of the next ξ unchanged, the joint law of $(D_1, \ldots, D_N)$ is unchanged. A no-contact step can therefore act only through the robot state the next placement starts from, which is what home vs. no reset measures.

Bias corollary. Replace ξ_k by $b + \xi_k$ in the proof of Proposition 2: the induction goes through whenever $|b + \xi_k| \le R - \tau$, and $D_k = \eta_k$ does not depend on b, so the bias is removed and the policy term becomes $\Pr(|b + \xi| > R - \tau)$.

Linear contraction. The measured contraction ratio fits a weaker, linear version of the funnel, $D_k = aD_{k-1} + c\,\xi_k$ with inheritance a and contraction c. For $|a| < 1$ and Gaussian ξ, $\mathrm{Var}(D_k) = c^2\sigma^2(1 - a^{2k})/(1 - a^2) \le c^2\sigma^2/(1 - a^2)$ for every k, so the error is bounded at every horizon, but still proportional to σ; the funnel model above is the limit in which the residual stops depending on the landing error, which is what the transfer sweeps test.